

Method	Cropped Word Recognition Accuracy(%)				
	IC03-50	IC03-FULL	SVT-50	IIIT5k-50	IIIT5k-1K
Wang et al. 2011	76.0	62.0	57.0	64.1	57.5
Mishra et al. 2012	81.8	67.8	73.2	-	-
Novikova et al. 2012	82.8	-	72.9	-	-
TSM+CRF(?)	87.4	79.3	73.5	-	-
Lee et al. 2014	88.0	76.0	80.0	-	-
Strokelets(?)	88.5	80.3	75.9	80.2	69.3
Wang et al. 2012	90.0	84.0	70.0	-	-
Alsharif and Pineau 2013	93.1	88.6	74.3	-	-
Su and Lu 2014	92.0	82.0	83.0	-	-
DeepFeatures	96.2	91.5	86.1	-	-
Goel et al. 2013	89.7	-	77.3	-	-
Almazán et al. 2014	-	-	87.0	88.6	75.6
DTRN	97.0	93.8	93.5	94.0	91.5
PhotoOCR	-	-	90.4	-	-
Jaderberg2015a	97.8	97.0	93.2	95.5	89.6
Jaderberg2015b	98.7	98.6	95.4	97.1	92.7

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Table 1: Cropped word recognition results on the SVT, ICDAR 2003, and IIIT 5K-word. The bottom figure shows unconstrained recognitions of the DTRN and the publicly available model (?), which is similar to *Jaderberg2015a*. Obviously, it seems to be sensitive to non-character spaces.

word recognition problem as a large-scale classification task by considering the images of a same word as a class. Thus the output layer should include a large number of classes, e.g. 90,000, imposing a huge number of model parameters which are difficult to be trained. Furthermore, it is not flexible to recognize a new word not trained. While the scene texts often include many irregular word strings (the number could be unlimited) which are impossible to be known in advanced, such as "AB00d". Thus our DTRN can process unknown words and arbitrary strings, providing a more flexible approach for this task.

Conclusion

We have presented a Deep-Text Recurrent Network (DTRN) for scene text recognition. It models the task as a deep sequence labelling problem that overcomes a number of main limitations. It computes a set of explicitly-ordered deep features from the word image, which is not only robust to low-level image distortions, but also highly discriminative to word strings. The recurrence property makes it capable of recognising highly ambiguous images by leveraging meaningful word context information, and also allows it to process unknown words and arbitrary strings, providing a more principled approach for this task. Experimental results show that our model has achieved the state-of-the-art performance.

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